Exercise – Understanding the theory of bootstrap

1. Aim of the exercise

In statistical analysis, assessing the reliability of a computed statistic is crucial. Ideally, this would involve evaluating its behavior across repeated samples from the true population distribution. However, in most practical scenarios, the full population is inaccessible. To address this limitation, we approximate the statistic's sampling distribution using resampling techniques. One powerful method for doing so is the bootstrap, which enables us to estimate uncertainty and variability without relying on strong parametric assumptions. In this exercise, we review the theoretical foundations of the bootstrap method.

2. Theory

The sampling distribution of a statistic can be characterized either by its cumulative distribution function (CDF) or, if the distribution is continuous, by its probability density function (PDF). Here, we focus on the CDF, as it directly conveys the probabilities required for statistical inference, such as the critical values of hypothesis tests and the quantiles used to construct confidence intervals.

Let $X_1, \ldots, X_n$ be a random sample of independent and identically distributed (i.i.d.) draws from the distribution of a random variable X, with cumulative distribution function (CDF)

$$F(x) \equiv P(X \leq x). \tag{1}$$

Let $T_n = T(X_1, \ldots, X_n)$ be a statistic computed from this sample, such as an estimator or a test statistic. Because T_n is a function of random variables, it is itself a random variable, and its distribution across repeated samples of size n is called its sampling distribution.

The CDF of T_n is

$$G_n(\tau, F) \equiv P_F(T_n \leq \tau). \tag{2}$$

The notation makes explicit that this CDF depends on the sample size n and on the distribution F from which the data are drawn.

When the distribution of T_n is continuous, its corresponding PDF is given by

$$g_n(\tau, F) = \frac{\partial G_n(\tau, F)}{\partial \tau}. \tag{3}$$

As an example, let $\mu = \mathbb{E}(X)$ and $\sigma^2 = \mathrm{Var}(X) < \infty$, and consider the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

We study it through the centred and scaled statistic

$$T_n = \sqrt{n}\,(\bar{X}_n - \mu). \tag{4}$$

The distribution of T_n has mean zero and variance σ^2 for every n. Unlike the distribution of $\bar{X}_n$ itself, whose variance σ^2/n tends to zero as n increases, the distribution of T_n does not collapse to a single point. This normalization therefore allows us to study the behavior of the sample mean as the sample size grows.

In practice, $G_n(\tau, F)$ is often unknown because F itself is rarely known and, even when F is specified, deriving $G_n(\tau, F)$ may be difficult or impossible. Conventional asymptotic theory addresses this problem by approximating $G_n(\tau, F)$ with its limiting form $G_\infty(\tau, F)$, which describes the distribution of T_n as $n \to \infty$. For many statistics, this limit follows from the Central Limit Theorem (CLT) or other asymptotic results and takes a known parametric form, such as the normal or chi-squared distribution. For the sample mean, the CLT implies that

$$G_\infty(\tau, F) = \Phi\Big(\frac{\tau}{\sigma}\Big),$$

where Φ denotes the standard normal CDF.

At first sight, this seems to solve the problem. However, the limiting distribution still depends on the unknown population distribution F, here through the unknown parameter σ. To make the approximation operational, the unknown population quantities must therefore be replaced by estimates computed from the sample. Typically, this is done through a parametric estimate $\hat{F} = F(\cdot, \hat{\theta})$, where $\hat{\theta}$ is a consistent estimator of the population parameter θ_0. For the sample mean, this amounts to replacing σ by the sample standard deviation.

Such plug-in approximations are often useful, but they can perform poorly in finite samples, especially when the sample size is small, when the underlying model assumptions are violated, or when the distribution of T_n is strongly skewed. The normal approximation for the sample mean, for example, is symmetric and therefore ignores any skewness present in the true sampling distribution. The bootstrap provides an alternative approach that approximates the finite-sample distribution $G_n(\tau, F)$ directly from the observed data.

2.1. The bootstrap principle

The bootstrap, introduced by Efron (1979), approximates $G_n(\tau, F)$ without parametric assumptions on F and without relying on the limit as $n \to \infty$. It replaces the unknown distribution F by the empirical distribution function (EDF)

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} I\{X_i \le x\}, \tag{5}$$

where $I\{\cdot\}$ denotes the indicator function, which equals one when the condition holds and zero otherwise. The EDF assigns to each value x the proportion of sample observations less than or equal to x. Equivalently, F_n is the discrete distribution that places probability $1/n$ on each observation in the sample.

The Glivenko–Cantelli theorem guarantees that the EDF converges uniformly to the true distribution F:

$$\sup_{x\in\mathbb{R}} |F_n(x) - F(x)| \xrightarrow{a.s.} 0 \qquad \text{as} \qquad n \to \infty. \tag{6}$$

This result implies that, as the sample size grows, the empirical distribution becomes uniformly close to the population distribution. Consequently, F_n can be viewed as a nonparametric estimate of the unknown distribution F.

Replacing F by F_n in the sampling distribution yields the bootstrap distribution

$$G_n(\tau, F_n) \equiv P_{F_n}(T_n^* \le \tau), \tag{7}$$

where T_n^* denotes the bootstrap analogue of the statistic T_n. Drawing from F_n amounts to drawing observations with replacement from the original sample. A sample generated in this way is called a bootstrap sample.

For the sample mean, the statistic of interest is

$$T_n = \sqrt{n}\left(\bar{X}_n - \mu\right). \tag{8}$$

In the bootstrap world, the empirical distribution F_n plays the role of the population distribution. Since the mean of F_n is $\bar{X}_n$, the bootstrap counterpart of T_n is

$$T_n^* = \sqrt{n}\left(\bar{X}_n^* - \bar{X}_n\right), \tag{9}$$

where $\bar{X}_n^*$ denotes the mean of the bootstrap sample.

The probability in $G_n(\tau, F_n)$ is taken over the resampling process, with the original sample held fixed. The bootstrap distribution therefore describes how the statistic would vary if repeated samples were drawn from the empirical distribution rather than from the unknown population distribution.

2.2. Approximating the bootstrap distribution

Although F_n is known, computing $G_n(\tau, F_n)$ exactly requires evaluating the statistic in every possible bootstrap sample. There are n^n equally likely ordered bootstrap samples, because each of the n resampling draws can take any of the n observed values. Even for moderate sample sizes, this number becomes enormous. For example, when $n = 20$, $20^{20} \approx 1.05 \times 10^{26}$. Exact computation is therefore infeasible except in very small samples.

Instead, $G_n(\tau, F_n)$ is approximated using Monte Carlo simulation. We draw B independent bootstrap samples, compute the bootstrap statistic $T_{n,b}^*$ in each bootstrap sample $b = 1, \ldots, B$, and estimate the bootstrap CDF by the proportion of bootstrap statistics that do not exceed τ:

$$\widehat{G}_n(\tau, F_n) = \frac{1}{B}\sum_{b=1}^{B} I\{T_{n,b}^* \le \tau\}. \tag{10}$$

We call $\widehat{G}_n(\tau, F_n)$ the simulated bootstrap CDF. It is the empirical distribution function of the B bootstrap statistics, in the same way that F_n is the empirical distribution function of the n observations.

Table 1 summarizes the three CDFs that play a role in bootstrap inference.

Table 1: The three CDFs in bootstrap inference

Name	Notation	Definition	Availability
Sampling CDF	$G_n(\tau, F)$	$P_F(T_n \le \tau)$	Unknown, because F is unknown
Bootstrap CDF	$G_n(\tau, F_n)$	$P_{F_n}(T_n^* \le \tau)$	Known in principle, but infeasible to compute exactly
Simulated bootstrap CDF	$\widehat{G}_n(\tau, F_n)$	$\frac{1}{B}\sum_{b=1}^{B} I\{T_{n,b}^* \le \tau\}$	Computed from B bootstrap samples

2.3. Two sources of error and two convergence results

In practice, the simulated bootstrap CDF $\widehat{G}_n(\tau, F_n)$ is used to approximate the sampling CDF $G_n(\tau, F)$. Their difference can be decomposed as

$$\widehat{G}_n(\tau, F_n) - G_n(\tau, F) = \underbrace{\widehat{G}_n(\tau, F_n) - G_n(\tau, F_n)}_{\text{simulation error}} + \underbrace{G_n(\tau, F_n) - G_n(\tau, F)}_{\text{statistical error}}. \tag{11}$$

The simulation error arises because the bootstrap distribution is approximated using a finite number of bootstrap samples B. The statistical error arises because the bootstrap draws from F_n rather than from the true population distribution F, which is the price of observing a finite sample of size n. Each source of error is controlled by its own convergence result.

2.3.1. Convergence as the number of bootstrap replications grows

Given the original sample, the indicators $I\{T^*_{n,b} \leq \tau\}$, $b = 1, \ldots, B$, are i.i.d. Bernoulli random variables with success probability $G_n(\tau, F_n)$. By the Law of Large Numbers, their average converges to its expectation:

$$\widehat{G}_n(\tau, F_n) \xrightarrow{p} G_n(\tau, F_n) \qquad \text{as} \qquad B \to \infty. \tag{12}$$

This result implies that the simulation error becomes arbitrarily small when the number of bootstrap samples is sufficiently large. Increasing B therefore improves the approximation of the bootstrap distribution, but it does not reduce the statistical error. As $B \to \infty$, $\widehat{G}_n(\tau, F_n)$ converges to $G_n(\tau, F_n)$ rather than to $G_n(\tau, F)$.

2.3.2. Convergence as the sample size grows

Since both $G_n(\tau, F_n)$ and $G_n(\tau, F)$ depend on the sample size, the second convergence result is stated in terms of their difference:

$$\sup_{\tau \in \mathbb{R}} |G_n(\tau, F_n) - G_n(\tau, F)| \xrightarrow{p} 0 \qquad \text{as} \qquad n \to \infty. \tag{13}$$

A bootstrap procedure satisfying this condition is called *consistent*. The intuition follows from the Glivenko–Cantelli theorem: as the sample size grows, the empirical distribution F_n becomes increasingly close to the true distribution F. Consequently, the sampling distribution generated from F_n becomes increasingly close to the sampling distribution generated from F.

Unlike the simulation error, the statistical error cannot be reduced by drawing more bootstrap samples. It can only be reduced by increasing the sample size.

Together, these two convergence results imply that the simulated bootstrap CDF $\widehat{G}_n(\tau, F_n)$ provides a good approximation to the sampling CDF $G_n(\tau, F)$ when both B and n are sufficiently large. In practice, the sample size n is determined by the available data, whereas the number of bootstrap samples B is chosen by the researcher. Next, we visualize both convergence results using simulation.

2.4. Visualizing the convergence results

We visualize the two convergence results using the sample mean. The data are drawn from an exponential distribution with rate $\lambda = 1/2$, so that $\mu = \sigma = 2$. This distribution is strongly right-skewed, making the finite-sample limitations of asymptotic approximations visible and

the convergence of the bootstrap easy to observe. Moreover, the sampling distribution of the statistic is known exactly. Since

$$\bar{X}_n \sim \text{Gamma}(n, n\lambda),$$

the sampling distribution of

$$T_n = \sqrt{n}(\bar{X}_n - \mu)$$

can be derived analytically. This allows the bootstrap distribution to be compared directly with the true sampling distribution.

All samples are taken from a single sequence of 1,000 observations, with a sample of size n consisting of the first n observations in that sequence. The PDFs we present are kernel density estimates. For the CDF comparisons, we show the largest vertical distance between the two CDFs over the grid of τ values. This distance is the simulation counterpart of the supremum appearing in the consistency condition

$$\sup_{\tau \in \mathbb{R}} |G_n(\tau, F_n) - G_n(\tau, F)| \xrightarrow{p} 0.$$

The MATLAB code used to generate the figure is presented at the end of this file.

2.4.1. Convergence as the number of bootstrap replications grows

Figures 1 fixes a sample of size $n = 20$ and shows the simulated bootstrap PDF and CDF based on $B = 10$, 100, and $10,000$ bootstrap samples. These simulated distributions are compared with the bootstrap distribution itself, approximated using $100,000$ bootstrap samples, and with the exact sampling distribution.

With $B = 10$, the simulated bootstrap CDF is a coarse step function. As B grows, the simulated bootstrap distributions approach the bootstrap distribution, and the largest distance between the two CDFs falls substantially. This illustrates the convergence result

$$\widehat{G}_n(\tau, F_n) \xrightarrow{p} G_n(\tau, F_n) \qquad \text{as} \qquad B \to \infty.$$

Increasing B therefore reduces the simulation error discussed in Section 2.3. However, it does not reduce the statistical error. Consequently, the gap between the bootstrap distribution and the sampling distribution remains visible in every panel. For this particular sample of 20 observations, the empirical distribution is less dispersed than the population distribution, causing the bootstrap distribution to be too narrow and only weakly right-skewed relative to the true sampling distribution.

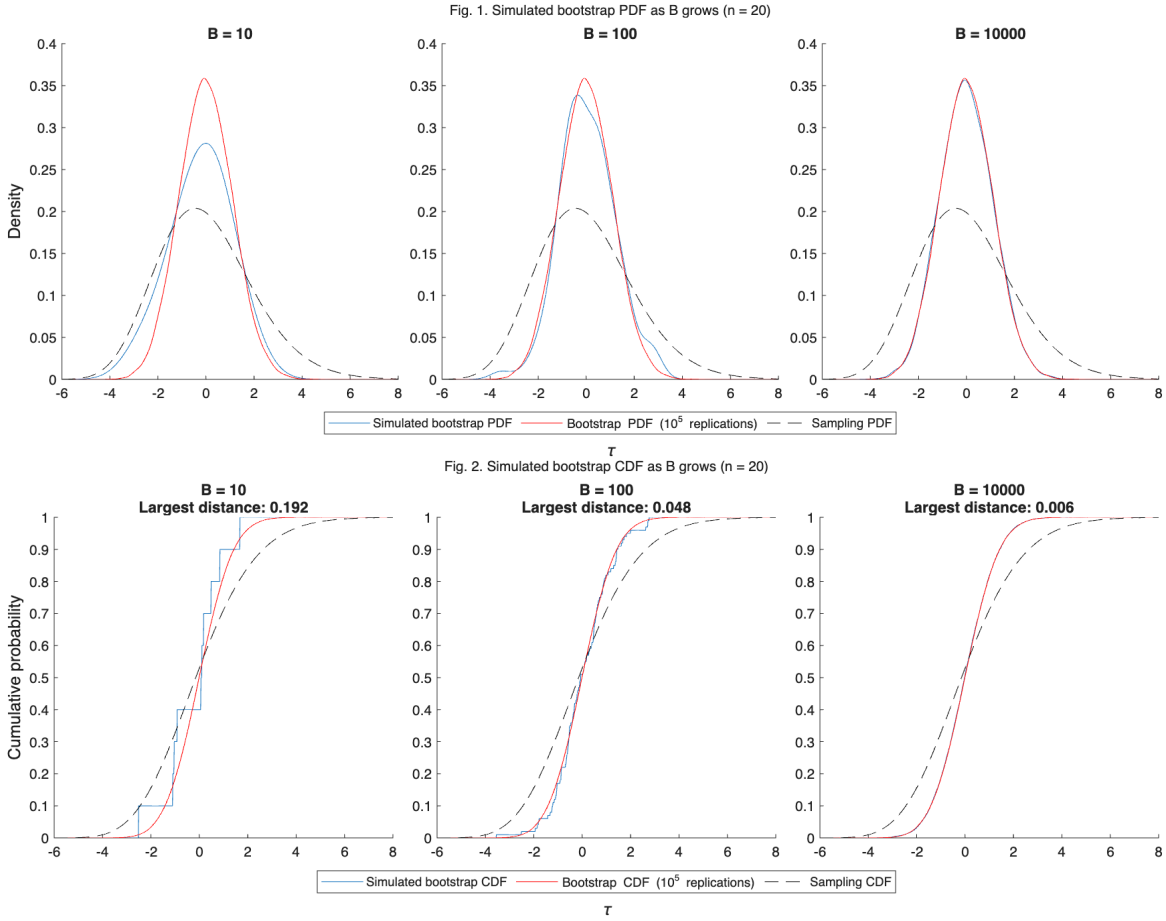


Figure 1: Simulated bootstrap PDF and CDF as the number of bootstrap samples increases while the sample size remains fixed.

2.4.2. Convergence as the sample size grows

Figure 2 increases the sample size from $n = 10$ to 100 and $1,000$, while fixing the number of bootstrap samples at $B = 10,000$. With such a large value of B, the simulation error is negligible, allowing the figures to focus on the statistical error.

For $n = 10$, the bootstrap distribution differs substantially from the sampling distribution. The bootstrap distribution is too narrow and nearly symmetric because a small sample from a skewed population often fails to capture the rare large observations that generate the skewness. As n increases, the empirical distribution F_n becomes increasingly similar to the population distribution F, causing the bootstrap and sampling distributions to move closer together. The largest distance between the two CDFs declines accordingly.

This illustrates the consistency result

$$\sup_{\tau \in \mathbb{R}} |G_n(\tau, F_n) - G_n(\tau, F)| \xrightarrow{p} 0 \qquad \text{as} \qquad n \to \infty.$$

Increasing the sample size therefore reduces the statistical error, whereas increasing the number of bootstrap samples only reduces the simulation error.

Together, the two figures illustrate the central message of bootstrap theory: a large value of B ensures that the bootstrap distribution is accurately simulated, while a large value of n ensures that the bootstrap distribution itself provides a good approximation to the true sampling distribution.

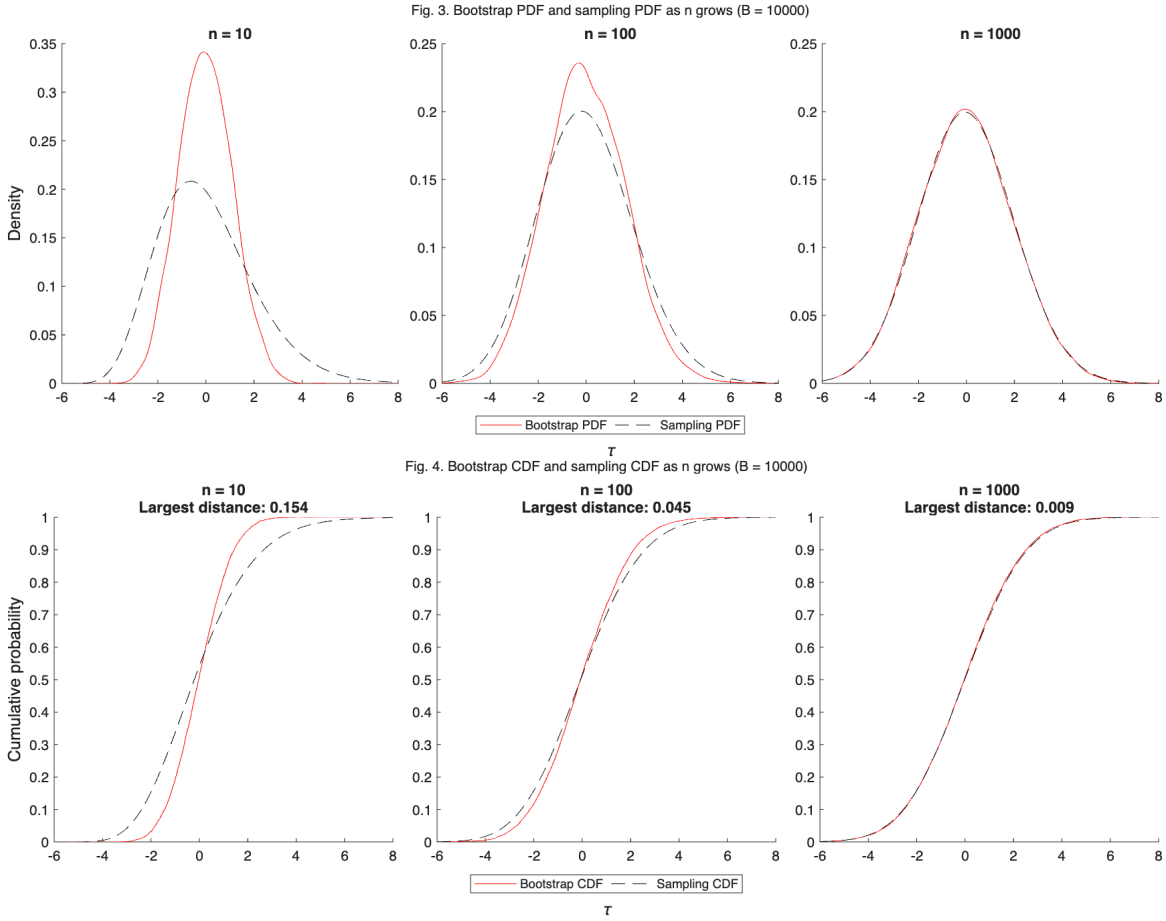


Figure 2: Bootstrap and sampling PDFs and CDFs as the sample size increases while the number of bootstrap samples remains fixed.

2.5. MATLAB code

The MATLAB code used to generate the figures is provided below.

```
%% 2. Initialize MATLAB environment

% Clear workspace, command window, and open figures
clear;
clc;
close all;

%% 3. Construct the data-generating process and analytical distributions

% 3.1. Set a seed for the random number generator for reproducibility
rng(26)

% 3.2. Set the mean of the exponential distribution generating the data
mu = 2;

% 3.3. Define the sampling PDF of the statistic
g_pop = @(tau,n) pdf('Gamma',mu + tau / sqrt(n),n,mu / n)/sqrt(n);

% 3.4. Define the sampling CDF of the statistic
G_pop = @(tau,n) cdf('Gamma',mu + tau / sqrt(n),n,mu / n);

% 3.5. Set the grid of tau values at which PDFs and CDFs are evaluated
tau_grid = linspace(-6,8,1000)';

% 3.6. Set the sample sizes
N_obs_vec = [10 100 1000];

% 3.7. Draw a sequence of observations from the population
data_all = random('Exponential',mu,[max(N_obs_vec),1]);

%% 4. Bootstrap distributions for different numbers of replications

% Prepare to study how the simulated bootstrap distribution converges to
% the bootstrap distribution as the number of bootstrap replications B
% increases.

% 4.1. Set the sample size
N_obs = 20;

% 4.2. Select the original sample
data_sample = data_all(1:N_obs);

% 4.3. Set the numbers of bootstrap replications to compare
N_boot_vec = [10 100 10000];

% 4.4. Set the benchmark number of bootstrap replications
N_boot_ref = 10^5;
```

```
61
62 % A large number of bootstrap replications is used to approximate the
63 % underlying bootstrap distribution.
64
65 % 4.5. Preallocate storage for the bootstrap statistic
66 T_boot = NaN(N_boot_ref,1);
67
68 % 4.6. Draw bootstrap samples and compute the bootstrap statistic
69 for b = 1:N_boot_ref
70     data_boot = datasample(data_sample,N_obs,'Replace',true);
71     % Tn* = sqrt(n) * (bootstrap sample mean - original sample mean)
72     T_boot(b) = sqrt(N_obs)*(mean(data_boot) - mean(data_sample));
73 end
74
75 % 4.7. Approximate the bootstrap PDF
76 g_boot_ref = ksdensity(T_boot,tau_grid,'function','pdf');
77
78 % 4.8. Approximate the bootstrap CDF
79 G_boot_ref = arrayfun(@(t) mean(T_boot <= t),tau_grid);
80
81 %% 5. Convergence from simulated bootstrap to bootstrap distribution
82
83 % 5.1. Plot the simulated bootstrap PDFs
84 figure('Position',[100 100 1200 450]);
85 layout = tiledlayout(1,3,'TileSpacing','compact','Padding','compact');
86 for i = 1:length(N_boot_vec)
87     B = N_boot_vec(i);
88     nexttile
89     hold on
90     plot(tau_grid,ksdensity(T_boot(1:B),tau_grid,'function','pdf'))
91     plot(tau_grid,g_boot_ref,'r')
92     plot(tau_grid,g_pop(tau_grid,N_obs),'k--')
93     title(['B = ' num2str(B)])
94     xlim([tau_grid(1) tau_grid(end)])
95     hold off
96 end
97 xlabel(layout,'\tau')
98 ylabel(layout,'Density')
99 lgd = legend( ...
100     'Simulated bootstrap PDF', ...
101     'Bootstrap PDF (10^5 replications)', ...
102     'Sampling PDF', ...
103     'Orientation','horizontal');
104 lgd.Layout.Tile = 'south';
105 t = title(layout, ...
106     ['Fig. 1. Simulated bootstrap PDF as B grows (n = ' ...
107     num2str(N_obs) ')']);
108 t.FontSize = get(groot,'defaultAxesFontSize');
109 t.FontWeight = 'normal';
110
```

```
% 5.2. Plot the simulated bootstrap CDFs
figure('Position',[100 100 1200 450]);
layout = tiledlayout(1,3,'TileSpacing','compact','Padding','compact');
for i = 1:length(N_boot_vec)
    B = N_boot_vec(i);
    G_boot = arrayfun(@(t) mean(T_boot(1:B) <= t),tau_grid);
    dist_max = max(abs(G_boot - G_boot_ref));
    nexttile
    hold on
    plot(tau_grid,G_boot)
    plot(tau_grid,G_boot_ref,'r')
    plot(tau_grid,G_pop(tau_grid,N_obs),'k--')
    title({['B = ' num2str(B)], ...
        ['Largest distance: ' num2str(dist_max,'%.3f')]})
    xlim([tau_grid(1) tau_grid(end)])
    hold off
end
xlabel(layout,'\tau')
ylabel(layout,'Cumulative probability')
lgd = legend( ...
    'Simulated bootstrap CDF', ...
    'Bootstrap CDF (10^5 replications)', ...
    'Sampling CDF', ...
    'Orientation','horizontal');
lgd.Layout.Tile = 'south';
t = title(layout, ...
    ['Fig. 2. Simulated bootstrap CDF as B grows (n = ' ...
    num2str(N_obs) ')']);
t.FontSize = get(groot,'defaultAxesFontSize');
t.FontWeight = 'normal';

%% 6. Bootstrap distributions for different sample sizes

% 6.1. Set the number of bootstrap replications
N_boot = 10^4;

% 6.2. Preallocate storage for the bootstrap statistic
T_boot_n = NaN(N_boot,length(N_obs_vec));

% 6.3. Compute the bootstrap statistic for each sample size
for i = 1:length(N_obs_vec)
    n = N_obs_vec(i);
    data_sample = data_all(1:n);
    for b = 1:N_boot
        data_boot = datasample(data_sample,n,'Replace',true);
        T_boot_n(b,i) = sqrt(n)*(mean(data_boot) - mean(data_sample));
    end
end

%% 7. Convergence from bootstrap distribution to sampling distribution
```

```
161
162 % 7.1. Plot the bootstrap and sampling PDFs
163 figure('Position',[100 100 1200 450]);
164 layout = tiledlayout(1,3, ...
165     'TileSpacing','compact', ...
166     'Padding','compact');
167 for i = 1:length(N_obs_vec)
168     n = N_obs_vec(i);
169     nexttile
170     hold on
171     plot(tau_grid,ksdensity(T_boot_n(:,i),tau_grid, ...
172         'function','pdf'),'r')
173     plot(tau_grid,g_pop(tau_grid,n),'k--')
174     title(['n = ' num2str(n)])
175     xlim([tau_grid(1) tau_grid(end)])
176     hold off
177 end
178 xlabel(layout,'\tau')
179 ylabel(layout,'Density')
180 lgd = legend( ...
181     'Bootstrap PDF', ...
182     'Sampling PDF', ...
183     'Orientation','horizontal');
184 lgd.Layout.Tile = 'south';
185 t = title(layout, ...
186     ['Fig. 3. Bootstrap PDF and sampling PDF as n grows (B = ' ...
187     num2str(N_boot) ')']);
188 t.FontSize = get(groot,'defaultAxesFontSize');
189 t.FontWeight = 'normal';
190
191 % 7.2. Plot the bootstrap and sampling CDFs
192 figure('Position',[100 100 1200 450]);
193 layout = tiledlayout(1,3, ...
194     'TileSpacing','compact', ...
195     'Padding','compact');
196 for i = 1:length(N_obs_vec)
197     n = N_obs_vec(i);
198     G_boot = arrayfun(@(t) mean(T_boot_n(:,i) <= t),tau_grid);
199     dist_max = max(abs(G_boot - G_pop(tau_grid,n)));
200     nexttile
201     hold on
202     plot(tau_grid,G_boot,'r')
203     plot(tau_grid,G_pop(tau_grid,n),'k--')
204     title({['n = ' num2str(n)], ...
205         ['Largest distance: ' num2str(dist_max,'%.3f')]})
206     xlim([tau_grid(1) tau_grid(end)])
207     hold off
208 end
209 xlabel(layout,'\tau')
210 ylabel(layout,'Cumulative probability')
```

```
lgd = legend( ...
    'Bootstrap CDF', ...
    'Sampling CDF', ...
    'Orientation','horizontal');
lgd.Layout.Tile = 'south';
t = title(layout, ...
    ['Fig. 4. Bootstrap CDF and sampling CDF as n grows (B = ' ...
    num2str(N_boot) ')']);
t.FontSize = get(groot,'defaultAxesFontSize');
t.FontWeight = 'normal';
```

3. Final notes

This file is prepared and copyrighted by Noortje Kamphuis and Tunga Kantarcı. This file and the accompanying MATLAB file are available on GitHub and can be accessed using this link.